MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY CASCADABLE, TTL, LATCHES, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, low-power Schottky TTL, latches. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	4-bit cascadable bistable latch
02	Quad cascadable $\overline{S}$ - $\overline{R}$ latch
03	8-bit cascadable addressable latch
04	4-bit cascadable bistable latch
05	8-bit cascadable addressable latch

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package
F	F-5 (16-lead, 1/4" x 3/8"), flat package
2	C-2 (20-terminal, .350" x .350"), square chip carrier package
X	C-2A (20-terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V to +7.0 V
Input voltage range	-1.5 V at -18 mA to +5.5 V
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) 1/:	
Device types 01 and 04	66 mW
Device type 02	38.5 mW
Device type 03	198 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case ($R_{\theta JC}$):	
Cases E, F, 2, and X	(See MIL-M-38510, appendix C)
Case F	70°C/W
Case 2	60°C/W
Junction temperature (T_J) 2/	175°C

1/ Must withstand added P_D due to short circuit test (e.g. I_{OS}).
2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC})	4.5 V minimum to 5.5 V maximum
Minimum high level input voltage (V_{IH})	2.0 V
Maximum low level input voltage (V_{IL})	0.7 V
Normalized fanout (each output)	10 maximum
Case operating temperature range (T_C)	-55°C to +125°C
Setup time, $t_{(SETUP)}$:	
Data to enable:	
Device types 01 and 04	20 ns minimum
Data to enable :	
Device type 03	17 ns minimum
Device type 05	24 ns minimum
Address to enable :	
Device type 03	15 ns minimum
Device type 05	0 ns minimum
Input hold time, $t_{(HOLD)}$:	
Data to enable:	
Device types 01 and 04	0 ns minimum
Data to enable :	
Device type 03	5 ns minimum
Device type 05	0 ns minimum
Address to enable :	
Device type 03	15 ns minimum
Device type 05	0 ns minimum

NOTE: Refers to rising or falling edge of the enable pulse.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

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3.2.1 Terminal connections. Terminal connections shall be as specified on figure 1.

3.2.2 Truth tables. The truth tables shall be as specified on figure 2.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 10 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007 as applicable of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 3, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in are optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Electrical test requirements shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 of table I of method 5005 of MIL-STD-883 shall be omitted.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, I_{OH} = -400 µA V_{IL} = 0.7 V, V_{IH} = 2.0 V	All	2.5	---	V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, I_{OL} = 4 mA V_{IL} = 0.7 V, V_{IH} = 2.0 V	All		0.4	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	All		-1.5	V
Low level input current	I_{IL1}	V_{CC} = 5.5 V, V_{IN} = 0.4 V				
At data			01	-.03	-0.42	mA
			04	-.0005	-0.4	mA
At $\overline{S}$, $\overline{R}$			02	0	-0.4	mA
At all inputs			03, 05	-.005	-.72	mA
Low-level input current, enable	I_{IL2}	V_{CC} = 5.5 V, V_{IN} = 0.4 V	01	-.06	-1.60	mA
			04	0	-1.2	mA
High level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.7 V				
At data			01, 04	---	20	µA
At $\overline{S}$, $\overline{R}$			02	---	20	µA
At all inputs			03, 05	---	20	µA
High-level input current, enable	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01, 04	---	80	µA
High level input current	I_{IH3}	V_{CC} = 5.5 V, V_{IN} = 5.5 V				
At data			01, 04	---	100	µA
At $\overline{S}$, $\overline{R}$			02	---	100	µA
At all inputs			03, 05	---	100	µA
High-level input current, enable	I_{IH4}	V_{CC} = 5.5 V, V_{IN} = 2.7 V	01, 04	---	400	µA

1/ Not more than one output should be shorted at a time.

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \le T_C \le +125°C$	Device type	Limits Min	Limits Max	Unit
Short-circuit output current	I_{OS}	V_{CC} = 5.5 V 1/	All	-15	-100	mA
Supply current	I_{CC}	V_{CC} = 5.5 V	01, 04	---	12	mA
			02	---	7	mA
			03, 05	---	36	mA
Low to high level, from D input to Q output	t_{PLH1}	C_L = 50 pF ±10%, R_L = 2 kΩ ±5%, V_{CC} = 5.0 V	01, 04	3	42	ns
High to low level, from D input to Q output	t_{PHL1}		01, 04	3	29	ns
Low to high level, from D input to $\overline{Q}$ output	t_{PLH2}		01, 04	3	32	ns
High to low level, from D input to $\overline{Q}$ output	t_{PHL2}		01, 04	3	26	ns
Low to high level, from enable input to Q output	t_{PLH3}		01, 04	3	42	ns
High to low level, from enable input to Q output	t_{PHL3}		01, 04	3	39	ns
Low to high level, from enable input to Q output	t_{PLH4}		01, 04	3	46	ns
High to low level, from enable input to Q output	t_{PHL4}		01, 04	3	26	ns
Low to high level, from $\overline{S}$ input to Q output	t_{PLH1}		02	3	35	ns
High to low level, from $\overline{S}$ input to Q output	t_{PHL1}		02	3	26	ns
High to low level, from $\overline{R}$ input to Q output	t_{PHL2}		02	3	42	ns
High to low level, from clear input to Q output	t_{PHL1}		03 05	3	42 30	ns
Low to high level, from data input to Q output	t_{PLH2}		03 05	3	48 46	ns

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TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits Min	Limits Max	Unit
High to low level, from data input to Q output	t_{PHL2}	C_L = 50 pF ±10%, R_L = 2 kΩ ±5% V_{CC} = 5.0 V	03 05	3	34 33	ns
Low to high level, from address input to Q output	t_{PLH3}		03 05	3	56 42	ns
High to low level, from address input to Q output	t_{PHL3}		03 05	3	44 42	ns
Low to high level, from enable input to Q output	t_{PLH4}		03 05	3	52 38	ns
High to low level, from enable input to Q output	t_{PHL4}		03, 05	3	38	ns

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) Class S devices	Subgroups (see table III) Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7,9,10,11	1*,2,3,7,9
Group A test requirements (method 5005)	1,2,3,7,8,9,10,11	1,2,3,7,8,9
Group B test requirements (method 5005) subgroup 5	1,2,3,9,10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

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Pin number		Pin name Device type 01		Pin name Device type 02		Pin name Device type 03		Pin name Device type 04		Pin name Device type 05	
Case											
2, X	E, F	2, X	E, F	2, X	E, F	2, X	E, F	2, X	E, F	2, X	E, F
1	1	NC	$1\overline{Q}$	NC	$1\overline{R}$	NC	A	NC	1D	NC	A
2	2	$1\overline{Q}$	1D	$1\overline{R}$	$1\overline{S}1$	A	B	1D	$1\overline{Q}$	A	B
3	3	1D	2D	$1\overline{S}1$	$1\overline{S}2$	B	C	$1\overline{Q}$	1Q	B	C
4	4	2D	ENLB 3-4	$1\overline{S}2$	1Q	C	Q0	1Q	ENLB 1-2	C	Q0
5	5	ENLB 3-4	V_{CC}	1Q	$2\overline{R}$	Q0	Q1	ENLB 1-2	2Q	Q0	Q1
6	6	NC	3D	NC	$2\overline{S}$	NC	Q2	NC	$2\overline{Q}$	NC	Q2
7	7	V_{CC}	4D	$2\overline{R}$	2Q	Q1	Q3	2Q	2D	Q1	Q3
8	8	3D	$4\overline{Q}$	$2\overline{S}$	GND	Q2	GND	2Q	GND	Q2	GND
9	9	4D	4Q	2Q	3Q	Q3	Q4	2D	3D	Q3	Q4
10	10	$4\overline{Q}$	3Q	GND	$3\overline{R}$	GND	Q5	GND	$3\overline{Q}$	GND	Q5
11	11	NC	$3\overline{Q}$	NC	$3\overline{S}1$	NC	Q6	NC	3Q	NC	Q6
12	12	4Q	GND	3Q	$3\overline{S}2$	Q4	Q7	3D	ENLB 3-4	Q4	Q7
13	13	3Q	ENLB 1-2	$3\overline{R}$	4Q	Q5	DATA IN	$3\overline{Q}$	4Q	Q5	DATA IN
14	14	$3\overline{Q}$	$2\overline{Q}$	$3\overline{S}1$	$4\overline{R}$	Q6	ENLB	3Q	$4\overline{Q}$	Q6	$\overline{ENLB}$
15	15	GND	2Q	$3\overline{S}2$	$4\overline{S}$	Q7	CLR	ENLB 3-4	4D	Q7	CLR
16	16	NC	1Q	NC	V_{CC}	NC	V_{CC}	NC	V_{CC}	NC	V_{CC}
17		ENLB 1-2		4Q		DATA IN		4Q		DATA IN	
18		$2\overline{Q}$		$4\overline{R}$		ENLB		$4\overline{Q}$		ENLB	
19		2Q		$4\overline{S}$		CLR		4D		CLR	
20		1Q		V_{CC}		V_{CC}		V_{CC}		V_{CC}	

FIGURE 1

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Device types 01 and 04

Inputs		Outputs	
D	enable	Q	$\overline{Q}$
L	H	L	H
H	H	H	L
X	L	Q_0	$\overline{Q}_0$

H = high level, L = low level, X = irrelevant
Q_0 = the level of Q before the high-to-low transition of enable

Device type 02

Inputs		Outputs
$\overline{S}$†	$\overline{R}$	Q
H	H	Q_0
L	H	H
H	L	L
L	L	H*

H = high level, L = low level, Q_0 = the level of Q before the indicated input conditions were established.
*This output level is pseudo stable; that is, it may not persist when the $\overline{S}$ and $\overline{R}$ inputs return to their inactive (hign) level.
† For latches with double $\overline{S}$ inputs:
H = both $\overline{S}$ inputs high
L = one or both $\overline{S}$ inputs low

FIGURE 2. Truth tables.

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Device types 03 and 05

Inputs		Output of addressed latch	Each other output	Function
Clear	enable			
H	L	D	Q_{i0}	Addressable latch
H	H	Q_{i0}	Q_{i0}	Memory
L	L	D	L	8-line demultiplexer
L	H	L	L	Clear

Latch Selection Table

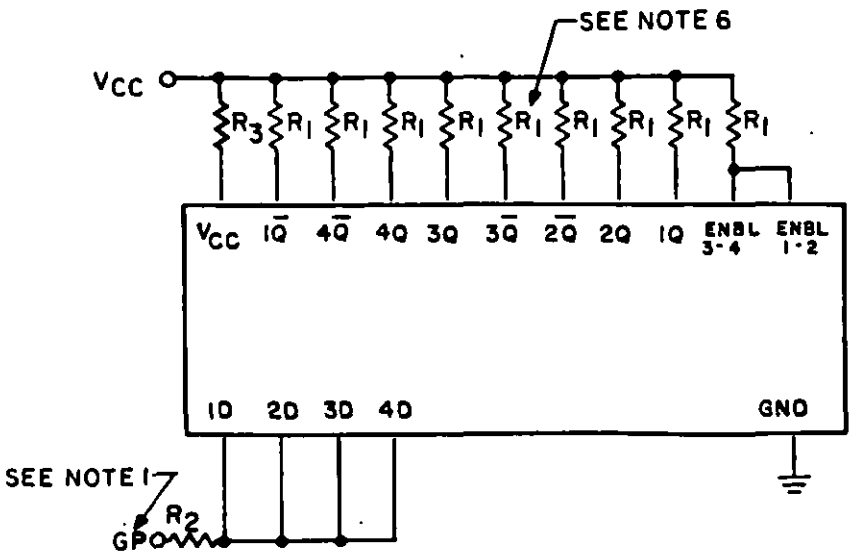
Select inputs			Latch addressed
C	B	A	
L	L	L	0
L	L	H	1
L	H	L	2
L	H	H	3
H	L	L	4
H	L	H	5
H	H	L	6
H	H	H	7

H = high level, L = low level
Q = the level at the data input
Q_{i0} = the level of Q_i (i = 0, 1,..., 7, as appropriate) before the indicated steady-state input conditions were established.

FIGURE 2. Truth tables - Continued.

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Device types 01 and 04



Device type 02

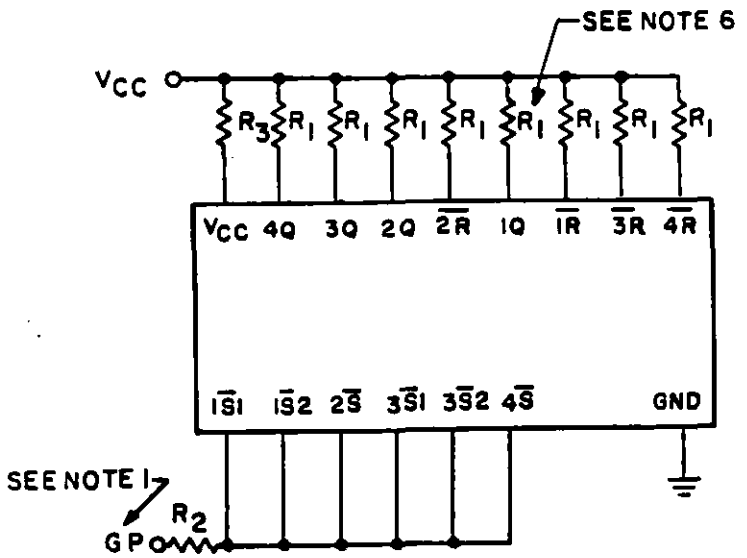
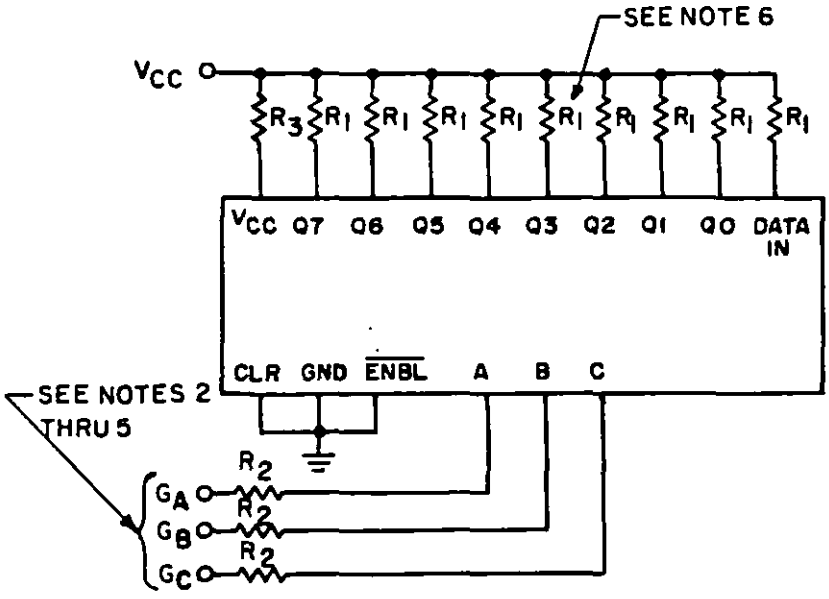


FIGURE 3. Burn-in and life test circuit.

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Device types 03 and 05



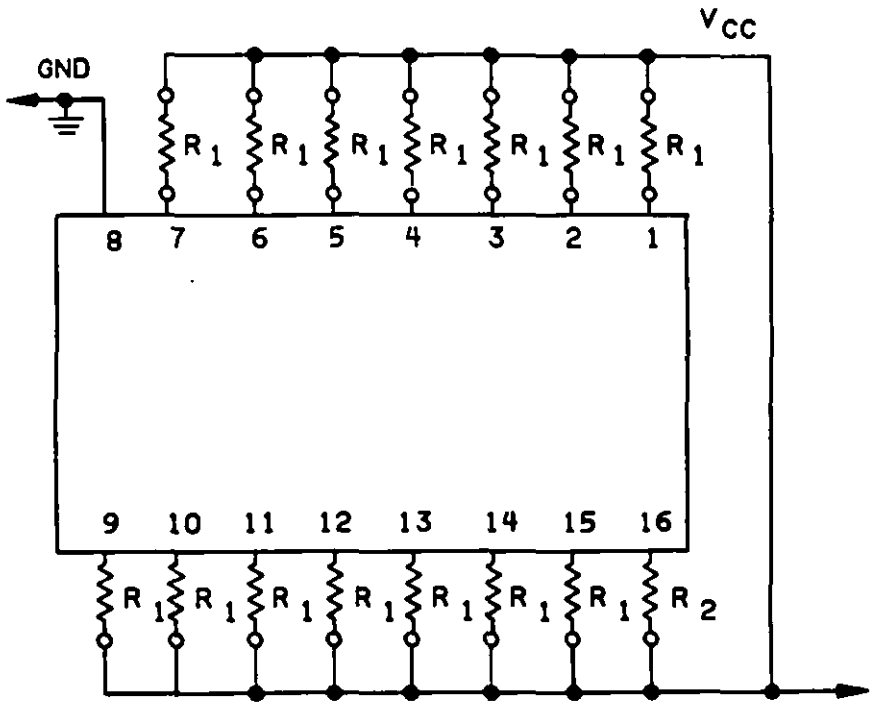
NOTES:
1. The GP pulse generator has the following characteristics: V_{IH} = 2.0 V minimum to 5.5 V maximum, V_{IL} = -0.5 V minimum to +0.7 V maximum, FREQ = 100 KHZ ±50 percent square wave, duty cycle = 50 percent ±15 percent.
2. The G_C pulse generator has the following characteristics: V_{IH} = 2.0 V minimum to 5.5 V maximum, V_{IL} = -0.5 V minimum to +0.7 V maximum, FREQ = 100 KHZ ±50 percent square wave, duty cycle = 50 percent ±15 percent.
3. The G_B pulse generator is identical to the G_C pulse generator except: FREQ = 2 X G_C.
4. The G_A pulse generator is identical to the G_C pulse generator except: FREQ = 4 X G_C.
5. The G_A, G_B and G_C pulse generators must be synchronized.
6. R_1 = 1 kΩ ±5 percent, R_2 = 27Ω maximum; R_3 = 10Ω maximum.
7. Voltage at V_{CC} terminal ≥ 5.0 V.

FIGURE 3. Burn-in and life test circuit - Continued.

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Device type 01

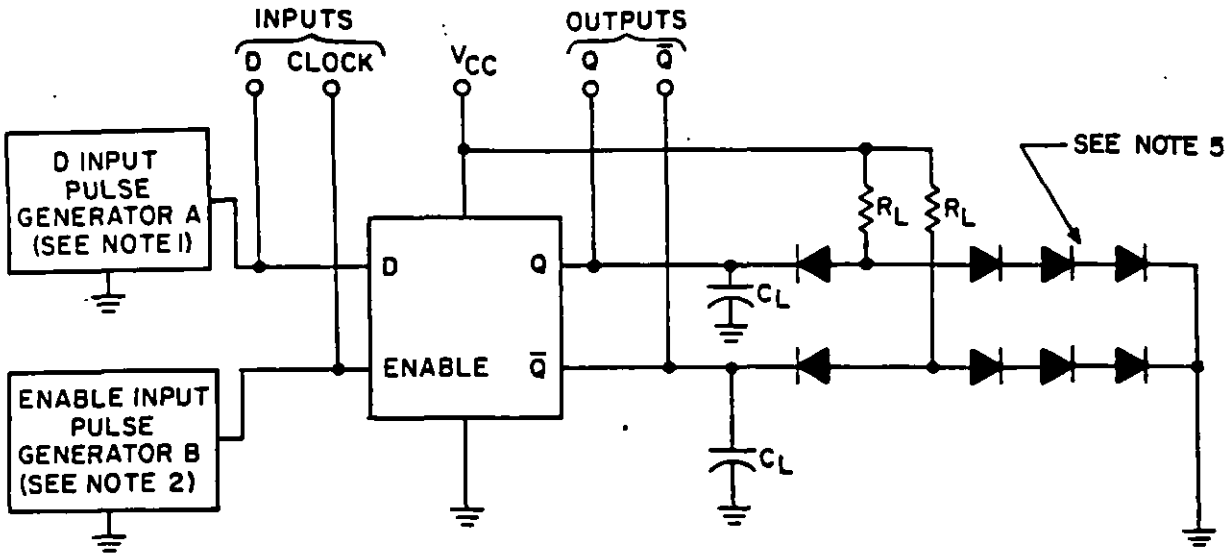
High temperature burn-in



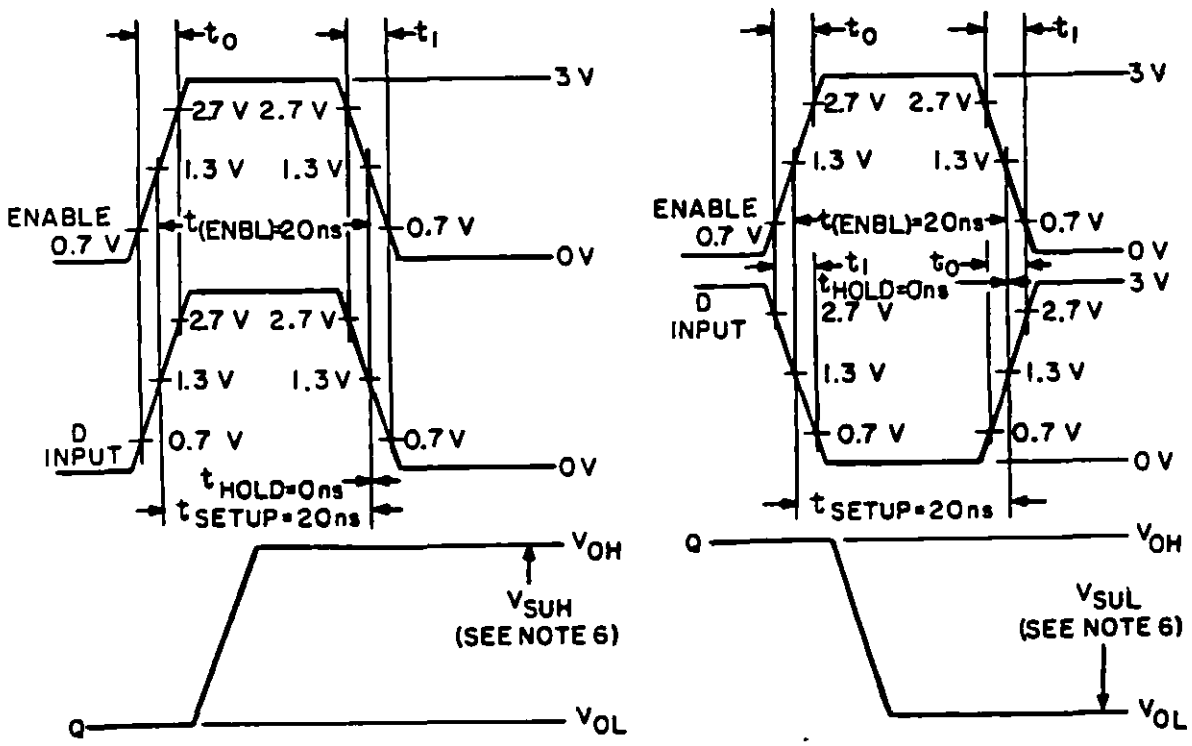
NOTES:
1. High temperature condition F(175°) condition F burn-in circuit only.
2. R_1 = 2 kΩ, R_2 = 2Ω.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuit - Continued.

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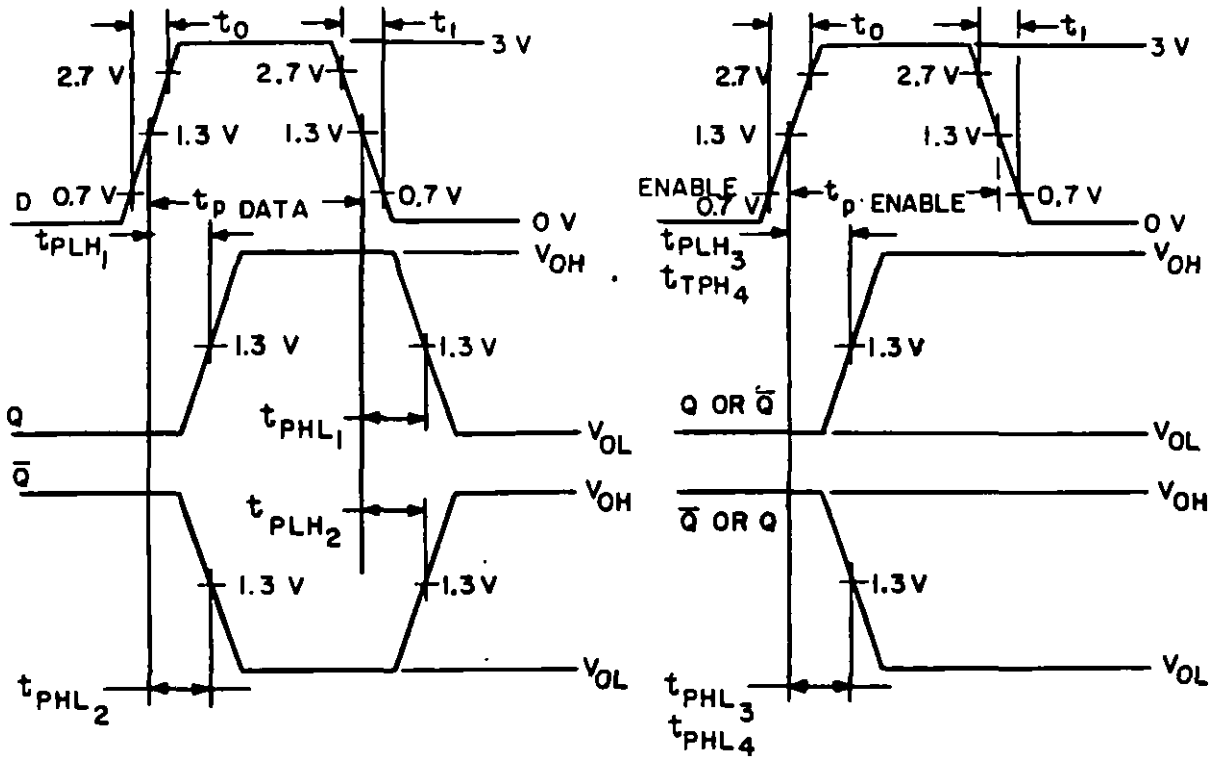
TEST CIRCUIT



SETUP AND HOLD WAVEFORMS

FIGURE 4. Switching test circuit and waveforms for device types 01 and 04.

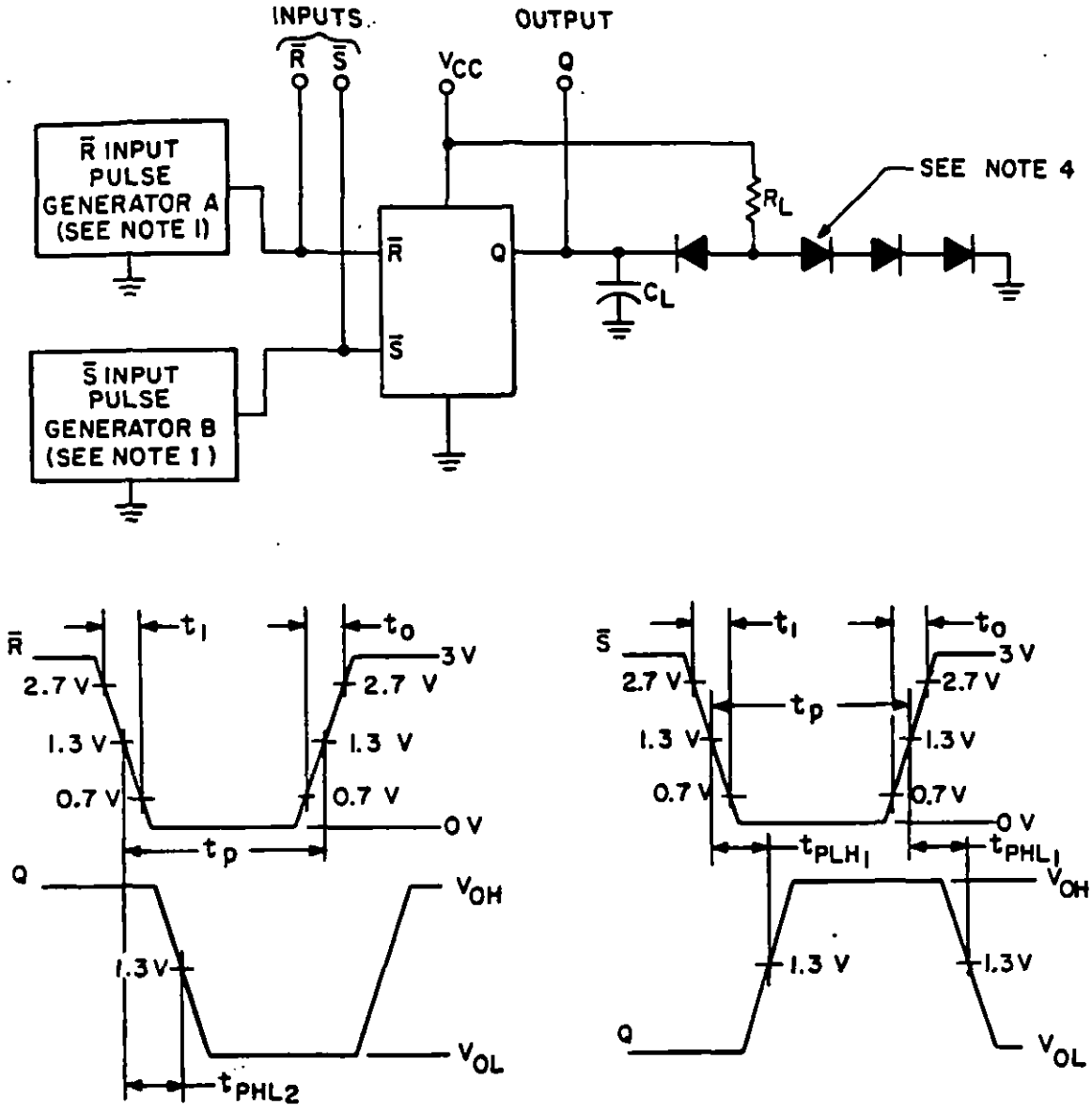
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NOTES:
1. The D input pulse generator has the following characteristics: V_{GEN} = 3 V, $t_0 \leq 15$ ns, $t_1 \leq 6$ ns, t_p = 30 ns and Z_{OUT} = 50Ω except when measuring V_{SETUP}.
2. The enable pulse generator is identical to the D input pulse generator.
3. C_L = 50 pF ±10 percent, which includes probe and jig capacitance.
4. R_L = 2 kΩ ±5 percent.
5. All diodes are 1N3064 of equivalent.
6. V_{SETUP} is to be measured 500 ns minimum after input transitions to assure that the device has latched with minimum setup and maximum hold conditions applied to inputs.

FIGURE 4. Switching test circuit and waveforms for device types 01 and 04 - Continued.

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NOTES:
1. $\overline{R}$ and $\overline{S}$ pulse generators have the following characteristics: t_p = 40 ±10 ns, $t_0 \leq 15$ ns, $t_1 \leq 6$ ns and PRR ≤ 1.0 MHz.
2. C_L = 50 pF ±10 percent, which includes probe and jig capacitance.
3. R_L = 2 kΩ ±5 percent.
4. All diodes are 1N3064 or equivalent.

FIGURE 5. Switching test circuit and waveforms for device type 02.

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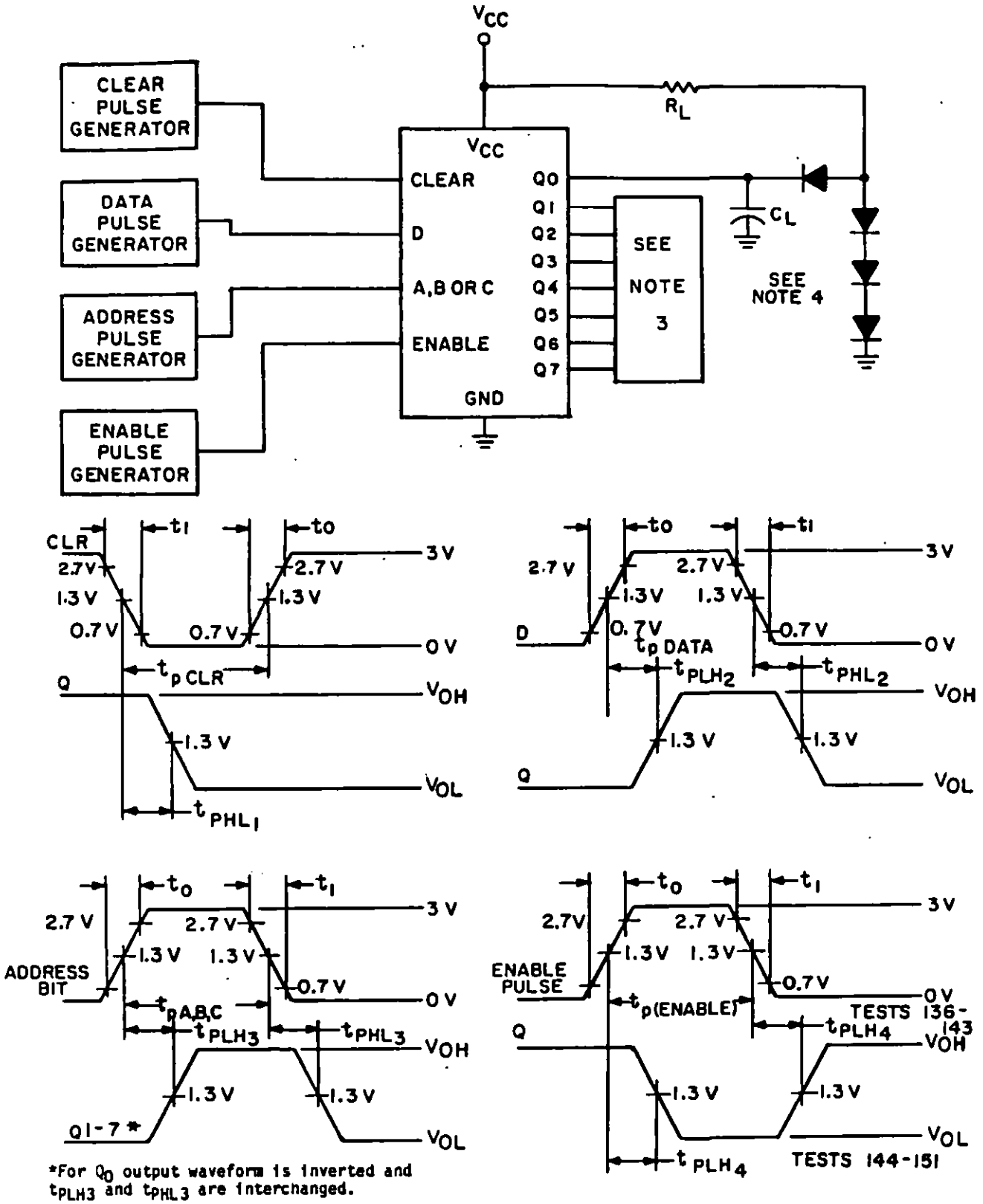
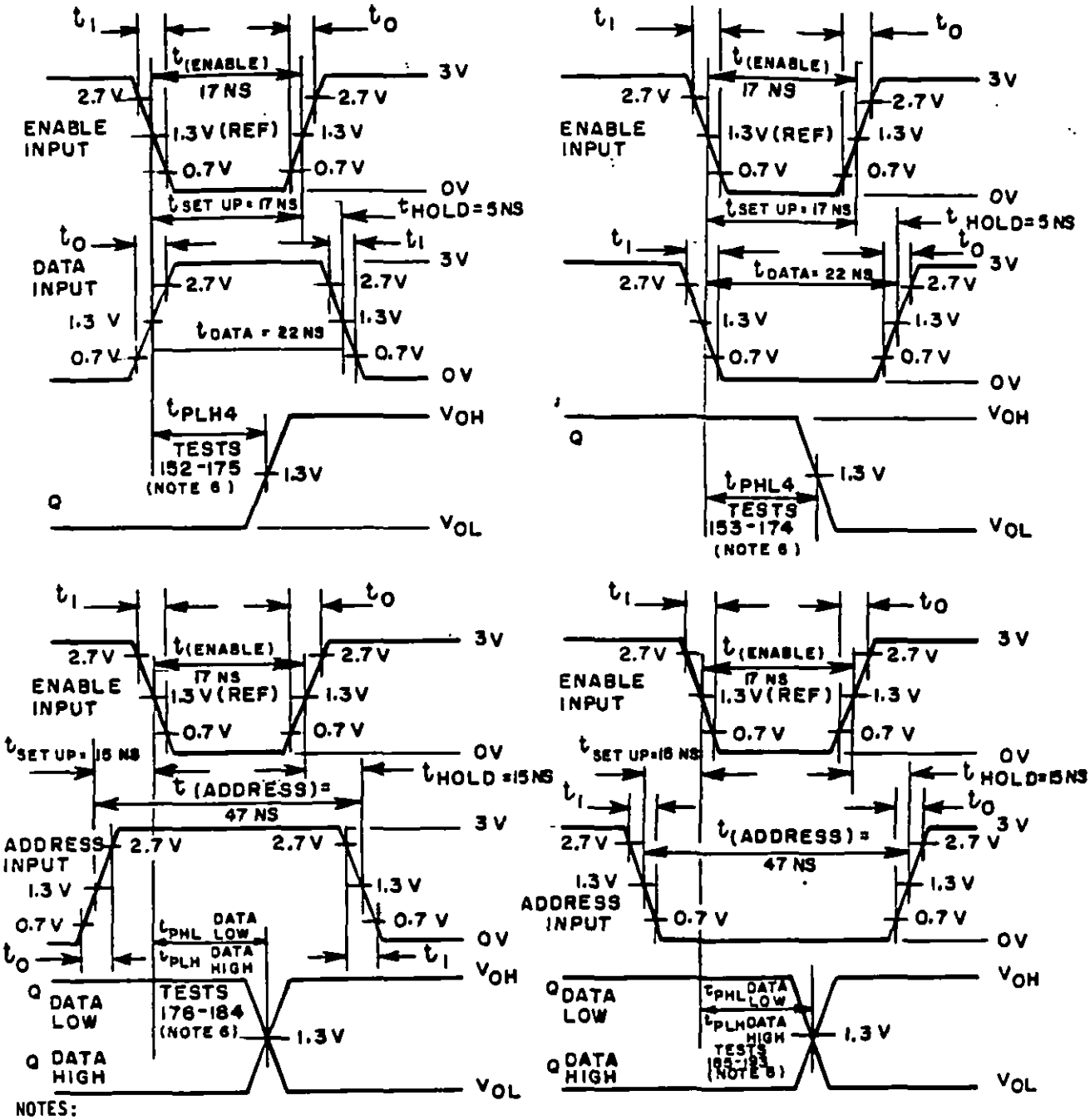


FIGURE 6. Switching test circuit and waveforms for device types 03 and 05.

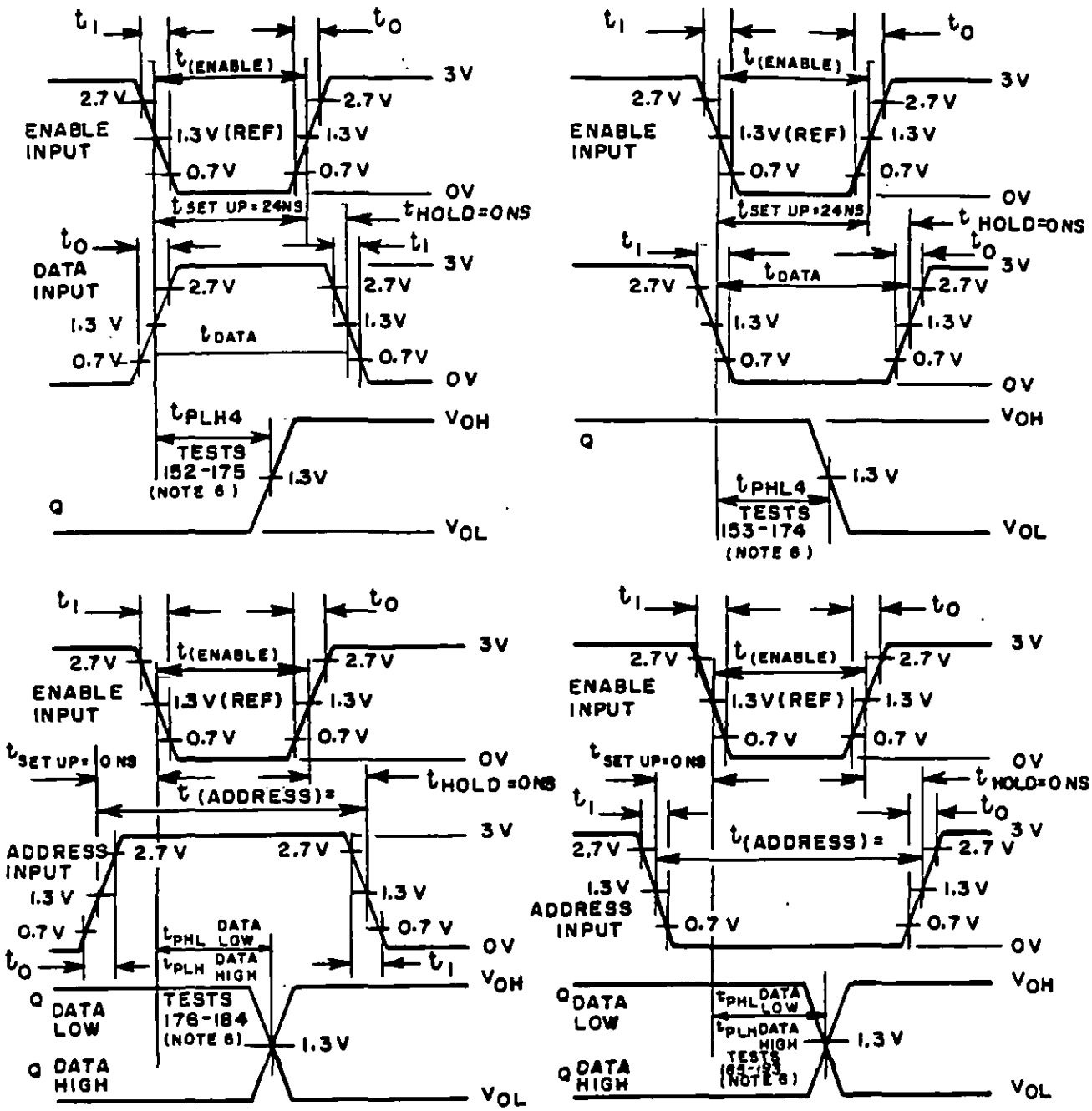
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NOTES:
1. R_L = 2 kΩ ±5 percent.
2. C_L = 50 pF ±10 percent including probe and jig capacitance.
3. All loads are the same as the Q_0 load.
4. All diodes are 1N3064 or equivalent.
5. The clear, enable, data, and address pulse generators have the following characteristics: V_{GEN} = 3 V, t_o ≤ 15 ns, t_l ≤ 6 ns, t_p = 30 ns, and PRR ≤ 1 MHz except when measuring test nos. 152 thru 193, t_p (ENABLE) = 17 ns, t_p (DATA) = 22 ns, t_p (ADDRESS) = 47 ns, and t_{SETUP} and t_{HOLD} are as specified on the waveforms above.
6. Immediately prior to test 152, all outputs shall be cleared low; then beginning with test 152, test 152 thru 193 are to be performed in sequence with a wait of 500 ns minimum between each test. These tests are to assure latchup of the outputs under worst case setup and hold input conditions.

FIGURE 6. Switching test circuit and waveforms for device type 03 - Continued.

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NOTES:
1. R_L = 2 kΩ ±5 percent.
2. C_L = 50 pF ±10 percent including probe and jig capacitance.
3. All loads are the same as the Q_0 load.
4. All diodes are 1N3064 or equivalent.
5. The clear, enable, data, and address pulse generators have the following characteristics: V_{GEN} = 3 V, t_l ≤ 6 ns, t_p = 30 ns, and PRR ≤ 1 MHz except when measuring test nos. 152 thru 193, t_p (ENABLE) = 24 ns, t_p (DATA) = 24 ns, t_p (ADDRESS) = 24 ns, and t_{SETUP} and t_{HOLD} are as specified on the waveforms above.
6. Immediately prior to test 152, all outputs shall be cleared low; then beginning with test 152, test 152 thru 193 are to be performed in sequence with a wait of 500 ns minimum between each test. These tests are to assure latchup of the outputs under worst case setup and hold input conditions.

FIGURE 6. Switching test circuit and waveforms for device type 05 - Continued.

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TABLE III. Group A inspection for device types 01 and 04.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Type 01 Cases E, F / Cases 2, X / Type 04 Cases E, F / Cases 2, X / Test no.	1 / 2 / 2 / 3 / 10	2 / 3 / 1 / 2 / 1D	3 / 4 / 7 / 8 / 2D	4 / 5 / 12 / 15 / EN 3-4	5 / 7 / 16 / 20 / V_{CC}	6 / 8 / 9 / 12 / 3D	7 / 9 / 15 / 19 / 4D	8 / 10 / 14 / 18 / 4$\overline{D}$	9 / 12 / 13 / 17 / 4Q	10 / 13 / 11 / 14 / 3Q	11 / 14 / 10 / 13 / 3$\overline{Q}$	12 / 15 / 8 / 10 / GND	13 / 17 / 4 / 5 / EN 1-2	14 / 18 / 6 / 8 / 2Q	15 / 19 / 5 / 7 / 2$\overline{Q}$	16 / 20 / 3 / 4 / 1Q	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1 2 3 4 5 6 7 8	-0.4 mA	0.7 V 2.0 V	0.7 V 2.0 V	1/	4.5 V	0.7 V 2.0 V	0.7 V 2.0 V	-0.4 mA	-0.4 mA	-0.4 mA	-0.4 mA	GND	1/	-0.4 mA	-0.4 mA	-0.4 mA	1Q 1$\overline{Q}$ 2Q 2$\overline{Q}$ 3Q 3$\overline{Q}$ 4Q 4$\overline{Q}$	2.5		V
	V_{OL}	3007	9 10 11 12 13 14 15 16	4 mA	2.0 V 0.7 V	2.0 V 0.7 V	1/		2.0 V 0.7 V	2.0 V 0.7 V	4 mA	4 mA	4 mA	4 mA	4 mA	1/	4 mA	4 mA	4 mA	1Q 1$\overline{Q}$ 2Q 2$\overline{Q}$ 3Q 3$\overline{Q}$ 4Q 4$\overline{Q}$		0.4	V
	I_{IL1}	3009	17 18 19 20		0.4 V	0.4 V	4.5 V 4.5 V	5.5 V	0.4 V	0.4 V						4.5 V 4.5 V				1D 2D 3D 4D	4/	4/	mA
	I_{IL2}		21 22		4.5 V	4.5 V	0.4 V		4.5 V	4.5 V						0.4 V				EN 1-2 EN 3-4			
	I_{IH1}	3010	23 24 25 26		2.7 V	2.7 V			2.7 V	2.7 V										1D 2D 3D 4D		20	μA
	I_{IH2}		27 28				2.7 V									2.7 V				EN 1-2 EN 3-4		80 80	
	I_{IH3}		29 30 31 32		5.5 V	5.5 V			5.5 V	5.5 V										1D 2D 3D 4D		100	
	I_{IH4}		33 34				5.5 V									5.5 V				EN 1-2 EN 3-4		0.4 0.4	mA
	I_{OS}	3011	35 36 37 38 39 40 41 42	GND	GND 4.5 V	GND 4.5 V	4.5 V		GND 4.5 V	GND 4.5 V	GND	GND	GND	GND	GND	4.5 V	GND	GND	GND	1D 1$\overline{D}$ 2D 2$\overline{D}$ 3D 3$\overline{D}$ 4D 4$\overline{D}$	-15	-100	
	I_{CC}	3005	43		GND	GND	GND		GND	GND						GND				V_{CC}		12	

See footnotes at end of device types 01 and 04.

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TABLE III. Group A inspection for device types 01 and 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Type 01 Cases E, F / Cases 2, X	Type 04 Cases E, F / Cases 2, X	Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			1 / 2	2 / 3		1Q	1D	2D	EN 3-4	V_{CC}	3D	4D	4Q	4Q	3Q	3Q	GND	EN 1-2	2Q	2Q	1Q				
1 T_C = +25°C	V_{IC}		44 45 46 47 48 49				18 mA	-18 mA	-18 mA		4.5 V " " " " "	-18 mA	-18 mA					GND " " " " "	-18 mA				1D 2D 3D 4D EN 1-2 EN 3-4		-1.5 " " " " "	V " " " " "
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																								
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																								
7 2/ T_C = +25°C	Truth table tests	3014	50 51 52 53		L " " H	A A B B	A A B B	A B B A	5.0 V " " "	A A B B	A A B B	L " " H	H " " L	H " " L	L " " H	GND " " "	A B B A	L " " H	H " " L	H " " L	3/ " " "			
8	Same tests and terminal conditions as for subgroup 7, except T_C = -55°C and +125°C.																								
9 T_C = +25°C	t_{PLH1} t_{PHL2} t_{PLH1} t_{PHL2} t_{PLH1} t_{PHL2} t_{PLH1} t_{PHL2} t_{PHL1} t_{PLH2} t_{PHL1} t_{PLH2} t_{PHL1} t_{PLH2}	3003 (Fig. 5)	54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69	OUT OUT	IN IN IN IN	IN IN IN IN	4.5 V 4.5 V 4.5 V	5.0 V	IN IN IN IN	IN IN IN IN	OUT OUT OUT OUT	OUT OUT OUT OUT	OUT OUT OUT OUT	GND 4.5 V OUT OUT	OUT OUT OUT OUT	OUT OUT OUT OUT	1D to 1Q 1D to 1Q 2D to 2Q 2D to 2Q 3D to 3Q 3D to 3Q 4D to 4Q 4D to 4Q 1D to 1Q 1D to 1Q 2D to 2Q 2D to 2Q 3D to 3Q 3D to 3Q 4D to 4Q 4D to 4Q	3 " " " " " " " " " " " " " " "	32 20 32 20 32 20 32 20 22 25 22 25 22 25 22 25	ns " " " " " " " " " " " " " " "
	t_{PLH3}		70 71 72 73		4.5 V	4.5 V	IN IN	4.5 V	IN IN	4.5 V	4.5 V	OUT	OUT	OUT	IN IN	OUT	1N to 1Q 1N to 2Q 1N to 3Q 1N to 4Q	"	32 " " "	" " " "
	t_{PHL3}		74 75 76 77		GND	GND	IN IN	GND	GND	GND	OUT	OUT	OUT	IN IN	OUT	OUT	1N to 1Q 1N to 2Q 1N to 3Q 1N to 4Q	"	30 " " "	" " " "
	t_{PLH4}		78 79 80 81	OUT	GND	GND	IN IN	GND	GND	OUT	OUT	OUT	IN IN	OUT	1N to 1Q 1N to 2Q 1N to 3Q 1N to 4Q	"	35 " " "	" " " "

See footnotes at end of device types 01 and 04.

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TABLE III. Group A inspection for device types 01 and 04 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
			Type 01 Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16				
			Cases 2, X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Type 04 Cases E, F	2	1	7	12	16	9	15	14	13	11	10	8	4	6	5	3				
			Cases 2, X	3	2	9	15	20	12	19	18	17	14	13	10	5	8	7	4				
			Test no.	1Q	1D	2D	EN 3-4	V_{CC}	3D	4D	4$\overline{Q}$	4Q	3Q	3$\overline{Q}$	GND	EN 1-2	2$\overline{Q}$	2Q	1Q				
9 T_C = +25°C	t_{PHL4}	3003 Fig. 5	82	OUT	4.5 V	4.5 V	IN	5.0 V	4.5 V	4.5 V	OUT			OUT	GND	IN	OUT			EN to 1Q	3	20	ns
		"	83				IN	"							"	IN				EN to 2Q	"	"	"
		"	84					"							"						EN to 3Q	"	"	"
		"	85					"							"						EN to 4Q	"	"	"
	V_{SUH}	"	86		IN	IN		"	IN				OUT		"	IN		OUT	OUT		1Q	2.5		V
		"	87				IN	"							"	IN					2Q	"		"
		"	88				IN	"		IN		OUT			"						3Q	"		"
		"	89					"							"						4Q	"		"
	V_{SUL}	"	90		IN	IN		"	IN				OUT		"	IN		OUT	OUT		1Q		0.4	"
		"	91				IN	"							"	IN					2Q		"	"
		"	92				IN	"		IN		OUT			"						3Q		"	"
		"	93					"							"						4Q		"	"
10 T_C=+125°C	t_{PLH1} t_{PHL1} t_{PLH2} t_{PHL2} t_{PLH3} t_{PHL3} t_{PLH4} t_{PHL4}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																			3	42	ns
																					"	29	"
																					"	32	"
																					"	26	"
																					"	42	"
																					"	39	"
																					"	46	"
																					"	26	"
	V_{SUH}																				2.5		V
	V_{SUL}																					0.4	V
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Apply 0 V/3 V - 5 V/0 V momentary pulse 500 ns minimum prior to measurement.

2/ A = 2.4 V, B = 0.4 V.

3/ H ≥ 1.5 V, L ≤ 1.5 V.

4/ I_{IL} limits are as follows:

Test	Min/max limits (mA)			
	Circuit A,B	Circuit C	Circuit D	Circuit E
			Device 01 / Device 04	
I_{IL1}	-.16/-0.4	-.0005/-.40	-.03/-.40 / -.16/-.40	-.19/-.42
I_{IL2}	-.64/-1.60	0/-1.2	-.12/-1.20 / -.64/-1.60	-.75/-1.6

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TABLE III. Group A inspection for device type 02.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases C, F / Cases 2, X Test no.	1 / 2	2 / 3	3 / 4	4 / 5	5 / 7	6 / 8	7 / 9	8 / 10	9 / 12	10 / 13	11 / 14	12 / 15	13 / 17	14 / 18	15 / 19	16 / 20	Measured terminal	Limits Min	Limits Max	Unit
				1R	1S1	1S2	1Q	2R	2S	2Q	GND	3Q	3R	3S1	3S2	4Q	4R	4S	V_{CC}				
1 T_C = +25°C	V_{OH}	3006	1 2 3 4	2.0 V	2.0 V	1/	-0.4 mA	2.0 V	1/	-0.4 mA	GND " "	-0.4 mA	2.0 V	2.0 V	1/	-0.4 mA	2.0 V	1/	4.5 V " "	4Q 3Q 2Q 1Q	2.5 " "		V " "
	V_{OL}	3007	5 6 7 8	1/	2.0 V	2.0 V	4 mA	1/	2.0 V	4 mA	" " "	4 mA	1/	2.0 V	2.0 V	4 mA	1/	2.0 V	" " "	1Q 2Q 3Q 4Q		0.4 " "	" " "
	I_{IL1}	3009	9 10 11 12 13 14 15 16 17 18	0.4 V GND GND	0.4 V 4.5 V	GND 4.5 V 0.4 V		0.4 V GND	GND 0.4 V		" " " " " " " " " "		0.4 V GND GND	0.4 V 4.5 V	GND 4.5 V 0.4 V		0.4 V GND	GND 0.4 V	5.5 V " " " " " " " " "	1R 1S1 1S2 2R 2S 3R 3S1 3S2 4R 4S	4/ " " " " " " " " "	4/ " " " " " " " " "	mA " " " " " " " " "
	I_{IH1}	3010	19 20 21 22 23 24 25 26 27 28	2.7 V	2.7 V	2.7 V		2.7 V	2.7 V		" " " " " " " " " "		2.7 V	2.7 V	2.7 V		2.7 V	2.7 V	" " " " " " " " " "	1R 1S1 1S2 2R 2S 3R 3S1 3S2 4R 4S		20 " " " " " " " " "	μA " " " " " " " " "
	I_{IH3}		29 30 31 32 33 34 35 36 37 38	5.5 V	5.5 V	5.5 V		5.5 V	5.5 V		" " " " " " " " " "		5.5 V	5.5 V	5.5 V		5.5 V	5.5 V	" " " " " " " " " "	1R 1S1 1S2 2R 2S 3R 3S1 3S2 4R 4S		100 " " " " " " " " "	" " " " " " " " " "
	I_{OS}	3011	39 40 41 42	5.0 V	GND	GND	GND	5.0 V	GND	GND	" " "	GND	5.0 V	GND	GND	GND	5.0 V	GND	" " " "	1Q 2Q 3Q 4Q	-15 " "	-100 " "	mA " "
	I_{CC}	3005	43	GND	4.5 V	4.5 V		GND	4.5 V		"		GND	4.5 V	4.5 V		GND	4.5 V	"	V_{CC}		7	"
	V_{IC}		44 45 46 47 48 49 50 51 52 53	-18 mA	-18 mA	-18 mA		-18 mA	-18 mA		" " " " " " " " " "		-18 mA	-18 mA	-18 mA		-18 mA	1m mA	4.5 V " " " " " " " " "	1R 1S1 1S2 2R 2S 3R 3S1 3S2 4R 4S		-1.5 " " " " " " " " "	V " " " " " " " " "

See footnotes at end of device type 02.

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TABLE III. Group A inspection for device type 07.

Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Cases 2, X Test no.	1 / 2 / IR	2 / 3 / 1S1	3 / 4 / 1S2	4 / 5 / 1Q	5 / 7 / 2R	6 / 8 / 2S	7 / 9 / 2Q	8 / 10 / GND	9 / 12 / 3Q	10 / 13 / 3R	11 / 14 / 3S1	12 / 15 / 3S2	13 / 17 / 4Q	14 / 18 / 4R	15 / 19 / 4S	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 2/ T_C = +25°C	Truth table tests	3014	54 55 56 57 58	B B A " "	B A A B A	B A " " B	H L L H H	B B A " "	B A A B B	H L L H H	GND " " " "	H L L H H	B B A " "	B A A B A	B A " " B	H L L H H	B B A " "	B A A B B	5.0 V " " " "	3/ " " " "			
8	Same tests and terminal conditions as for subgroup 7, except T_C = -55°C and +125°C.																						
9 T_C = +25°C	t_{PLH1}	3003 (Fig. 6) " " " "	59 60 61 62 63 64	GND GND	IN 4.5 V	4.5 V IN	OUT OUT	GND	IN	OUT	GND " " " " "	OUT OUT	GND GND	IN 4.5 V	4.5 V IN	OUT	GND	IN	5.0 V " " " "	1S1 to 1Q 1S2 to 1Q 2S to 2Q 3S1 to 3Q 3S2 to 3Q 4S to 4Q	3 " " " "	27 " " " "	ns " " " "
	t_{PHL1}	" " " " " "	65 66 67 68 69 70	GND GND	IN 4.5 V	4.5 V IN	OUT OUT	GND	IN	OUT	" " " " " "	OUT OUT	GND GND	IN 4.5 V	4.5 V IN	OUT	GND	IN	" " " " " "	1S1 to 1Q 1S2 to 1Q 2S to 2Q 3S1 to 3Q 3S2 to 3Q 4S to 4Q	" " " " " "	20 " " " " "	" " " " " "
	t_{PHL2}	" " " " "	71 72 73 74	IN	4.5 V	4.5 V	OUT	IN	4.5 V	OUT	" " " " "	OUT	IN	4.5 V	4.5 V	OUT	" IN	4.5 V	" " " " "	1R to 1Q 2R to 2Q 3R to 3Q 4R to 4Q	" " " " "	32 " " " "	" " " " "
10	t_{PLH1} t_{PHL1} t_{PHL2}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C, and limits as shown.																			" " "	35 26 42	" " "
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

1/ Apply a 3V-5V / 0 V momentary pulse 500 ns minimum prior to measurement.

2/ A = 2.4 V, B = 0.4 V.

3/ H ≥ 1.5 V, L ≤ 1.5 V.

4/ I_{IL} limits are as follows:

Symbol	Min/max limits (mA)							
	Circuit A		Circuit B	Circuit C	Circuit D		Circuit E
	Tests	Limits			Tests	Limits	
I_{IL}	9,12, 14,17	0/0.2	-.16/-.4	-.135/-.370	9,12, 14,17	-.001/-.15	-.16/-.4
	10,11, 13,15, 16,18	0/0.2			10,11, 13,15, 16,18	-.03/-.30	

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TABLE III. Group A inspection for device type 03.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Cases 2, x Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
				A	B	C	Q0	Q1	Q2	Q3	GND	Q4	Q5	Q6	Q7	DATA IN	ENBL	CLR	V_{LL}				
1 T_C = +25°C	V_{OH}	3006	1	0.7 V	0.7 V	0.7 V	-0.4 mA	-0.4 mA			GND					2.0 V	1/	2.0 V	4.5 V	Q0	2.5		V
			2	2.0 V	0.7 V	"			-0.4 mA							"	"	"	"	Q1	"		"
			3	0.7 V	2.0 V					-0.4 mA										Q2			
			4	2.0 V	2.0 V							-0.4 mA								Q3			
			5	0.7 V	0.7 V	2.0 V							-0.4 mA							Q4			
			6	2.0 V	0.7 V									-0.4 mA						Q5			
			7	0.7 V	2.0 V										-0.4 mA					Q6			
			8	2.0 V	2.0 V															Q7			
	V_{OL}	3007	9	0.7 V	0.7 V	0.7 V	4 mA	4 mA								0.7 V				Q0		0.4	V
			10	2.0 V	0.7 V	"			4 mA							"				Q1		"	"
			11	0.7 V	2.0 V					4 mA										Q2			
			12	2.0 V	2.0 V							4 mA								Q3			
			13	0.7 V	0.7 V	2.0 V							4 mA							Q4			
			14	2.0 V	0.7 V									4 mA						Q5			
			15	0.7 V	2.0 V										4 mA					Q6			
			16	2.0 V	2.0 V															Q7			
	I_{IL1}	3009	17	0.4 V	0.4 V	0.4 V										0.4 V	0.4 V	0.4 V	5.5 V	A	4/	4/	mA
			18																	B			
			19																	C			
			20																	D1			
			21																	ENBL			
			22																	CLR			
	I_{IH1}	3010	23	2.7 V	2.7 V	2.7 V										2.7 V	2.7 V	2.7 V		A		20	μA
			24																	B			
			25																	C			
			26																	D1			
			27																	ENBL			
			28																	CLR			
	I_{IH3}		29	5.5 V	5.5 V	5.5 V										5.5 V	5.5 V	5.5 V		A		100	μA
			30																	B			
			31																	C			
			32																	D1			
			33																	ENBL			
			34																	CLR			
	V_{IC}		35	-18 mA	-18 mA	-18 mA										-18 mA	-18 mA	-18 mA	4.5 V	A		-1.5	V
			36																	B			
			37																	C			
			38																	D1			
			39																	ENBL			
			40																	CLR			
	I_{OS}	3011	41	GND	GND	GND	GND	GND								5.0 V	GND	5.0 V	5.5 V	Q0	-15	-100	mA
			42	5.0 V	GND	"			GND							"	"	"	"	Q1	"	"	"
			43	GND	5.0 V					GND										Q2			
			44	5.0 V	5.0 V							GND								Q3			
			45	GND	GND	5.0 V							GND							Q4			
			46	5.0 V	GND	"								GND						Q5			
			47	GND	5.0 V										GND					Q6			
			48	5.0 V	5.0 V															Q7			
	I_{CC}	3005	49	GND	GND	GND										GND		GND		V_{CC}		36	mA

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases 2, X / Test no.	1 / 2 / A	2 / 3 / B	3 / 4 / C	4 / 5 / Q0	5 / 7 / Q1	6 / 8 / Q2	7 / 9 / Q3	8 / 10 / GND	9 / 12 / Q4	10 / 13 / Q5	11 / 14 / Q6	12 / 15 / Q7	13 / 17 / DATA IN	14 / 18 / ENBL	15 / 19 / CLR	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																					
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																					
7 2/ T_C = +25°C	Truth table tests	3014	50	B	B	B	L	L	L	L	GND	L	L	L	L	A	A	B	5.0 V	3/		
			51	B	-	-	H	-	-	-	-	-	-	-	-	-	B	-	-			
			52	A	-	-	L	-	-	-	-	-	-	-	-	-	A	-	-			
			53	A	-	-	-	H	-	-	-	-	-	-	-	-	B	-	-			
			54	B	A	-	-	L	-	-	-	-	-	-	-	-	A	-	-			
			55	B	-	-	-	-	H	-	-	-	-	-	-	-	B	-	-			
			56	A	-	-	-	-	L	-	-	-	-	-	-	-	A	-	-			
			57	A	-	-	-	-	-	H	-	-	-	-	-	-	B	-	-			
			58	B	B	-	-	-	-	L	-	-	-	-	-	-	A	-	-			
			59	B	-	A	-	-	-	-	-	H	-	-	-	-	B	-	-			
			60	A	-	-	-	-	-	-	-	L	-	-	-	-	A	-	-			
			61	A	-	-	-	-	-	-	-	-	H	-	-	-	B	-	-			
			62	B	A	-	-	-	-	-	-	-	L	-	-	-	A	-	-			
			63	B	-	-	-	-	-	-	-	-	-	H	-	-	B	-	-			
			64	A	-	-	-	-	-	-	-	-	-	L	-	-	A	-	-			
			65	-	-	-	-	-	-	-	-	-	-	-	H	-	B	-	-			
			66	B	-	-	-	-	-	-	-	-	-	-	-	-	A	A	-			
			67	A	B	-	-	-	-	-	-	-	-	-	-	B	-	-	-			
			68	A	A	B	-	-	-	-	-	-	-	-	-	-	B	-	-			
			69	B	B	A	-	-	-	-	-	-	-	-	-	-	A	-	-			
			70	A	A	-	B	-	-	-	-	-	-	-	-	-	B	-	-			
			71	B	A	A	-	-	-	-	-	-	-	-	-	-	A	-	-			
			72	A	B	-	-	-	-	-	-	-	-	-	-	A	-	-	-			
			73	B	-	B	-	-	-	-	-	-	-	-	-	-	A	-	-			
			74	-	-	-	H	-	-	-	-	-	-	-	-	-	B	-	-			
			75	A	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			76	-	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			77	B	A	-	-	H	-	-	-	-	-	-	-	-	B	-	-			
			78	-	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			79	B	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			80	-	-	-	-	-	H	-	-	-	-	-	-	-	A	-	-			
			81	A	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			82	-	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			83	B	B	A	-	-	-	H	-	-	-	-	-	-	B	-	-			
			84	-	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			85	B	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			86	-	-	-	-	-	-	-	-	H	-	-	-	-	A	-	-			
			87	A	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			88	-	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			89	B	A	-	-	-	-	-	-	-	H	-	-	-	A	-	-			
			90	-	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			91	B	-	-	-	-	-	-	-	-	-	H	-	-	A	-	-			
			92	-	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			93	A	-	-	-	-	-	-	-	-	-	-	H	-	A	-	-			
			94	-	-	-	-	-	-	-	-	-	-	-	-	-	A	-	-			
			95	A	-	-	-	-	-	-	-	-	-	-	-	-	B	-	-			
8	Same tests and terminal conditions as for subgroup 7, except T_C = -55°C and +125°C.																					

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Cases 2, X Test no.	1 / 2 A	2 / 3 B	3 / 4 C	4 / 5 Q0	5 / 7 Q1	6 / 8 Q2	7 / 9 Q3	8 / 10 GND	9 / 12 Q4	10 / 13 Q5	11 / 14 Q6	12 / 15 Q7	13 / 17 DATA IN	14 / 18 ENBL	15 / 19 CLR	16 / 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PHL1}	3003 Fig. 7	96	GND	GND	GND	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	4.5 V	6/	IN	5.0 V	CLR to Q0	3	32	ns
			97	4.5 V	GND															CLR to Q1			
			98	GND	4.5 V															CLR to Q2			
			99	4.5 V	4.5 V															CLR to Q3			
			100	GND	GND	4.5 V														CLR to Q4			
			101	4.5 V	GND															CLR to Q5			
			102	GND	4.5 V															CLR to Q6			
			103	4.5 V	4.5 V															CLR to Q7			
	t_{PLH2}		104	GND	GND	GND	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	IN	GND	GND		D1 to Q0	3	37	
			105	4.5 V	GND															D1 to Q1			
			106	GND	4.5 V															D1 to Q2			
			107	4.5 V	4.5 V															D1 to Q3			
			108	GND	GND	4.5 V														D1 to Q4			
			109	4.5 V	GND															D1 to Q5			
			110	GND	4.5 V															D1 to Q6			
			111	4.5 V	4.5 V															D1 to Q7			
	t_{PHL2}		112	GND	GND	GND	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT					D1 to Q0		26	
			113	4.5 V	GND															D1 to Q1			
			114	GND	4.5 V															D1 to Q2			
			115	4.5 V	4.5 V															D1 to Q3			
			116	GND	GND	4.5 V														D1 to Q4			
			117	4.5 V	GND															D1 to Q5			
			118	GND	4.5 V															D1 to Q6			
			119	4.5 V	4.5 V															D1 to Q7			
	t_{PLH3}		120	IN	GND	GND	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT	4.5 V				A to Q0		43	
			121	IN	GND															A to Q1			
			122	GND	IN															B to Q2			
			123	IN	4.5 V															A to Q3			
			124	GND	GND	IN														C to Q4			
			125	IN	GND	4.5 V														A to Q5			
			126	GND	IN															B to Q6			
			127	IN	4.5 V															A to Q7			
	t_{PHL3}		128	IN	GND	GND	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT					A to Q0		34	
			129	IN	GND															A to Q1			
			130	GND	IN															B to Q2			
			131	IN	4.5 V															A to Q3			
			132	GND	GND	IN														C to Q4			
			133	IN	GND	4.5 V														A to Q5			
			134	GND	IN															B to Q6			
			135	IN	4.5 V															A to Q7			
	t_{PLH4}		136	GND	GND	GND	OUT	OUT	OUT	OUT		OUT	OUT	OUT	OUT		IN			ENBL to Q0		39	
			137	4.5 V	GND															ENBL to Q1			
			138	GND	4.5 V															ENBL to Q2			
			139	4.5 V	4.5 V															ENBL to Q3			
			140	GND	GND	4.5 V														ENBL to Q4			
			141	4.5 V	GND															ENBL to Q5			
			142	GND	4.5 V															ENBL to Q6			
			143	4.5 V	4.5 V															ENBL to Q7			

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 03 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F Cases Z, K Test no.	1 / 2 / A	2 / 3 / B	3 / 4 / C	4 / 5 / Q0	5 / 7 / Q1	6 / 8 / Q2	7 / 9 / Q3	8 / 10 / GND	9 / 12 / Q4	10 / 13 / Q5	11 / 14 / Q6	12 / 15 / Q7	13 / 17 / DATA IN	14 / 18 / ENBL	15 / 19 / CLR	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PHL4}	3003 Fig. 7	144	GND	GND	GND	OUT				GND					4.5 V	IN	GND	5.0 V	ENBL to Q0	3	29	ns
			145	4.5 V	GND	"		OUT			"					"	"	"	"	ENBL to Q1	"	"	"
			146	GND	4.5 V	"			OUT		"					"	"	"	"	ENBL to Q2	"	"	"
			147	4.5 V	4.5 V	"				OUT	"					"	"	"	"	ENBL to Q3	"	"	"
			148	GND	GND	4.5 V					"	OUT				"	"	"	"	ENBL to Q4	"	"	"
			149	4.5 V	GND	"					"		OUT			"	"	"	"	ENBL to Q5	"	"	"
			150	GND	4.5 V	"					"			OUT		"	"	"	"	ENBL to Q6	"	"	"
			151	4.5 V	4.5 V	"					"				OUT	"	"	"	"	ENBL to Q7	"	"	"
	t_{PLH4}	*/	152	GND	GND	GND	OUT				"					IN	IN	4.5 V	"	ENBL to Q0	"	39	"
	t_{PHL4}	"	153	"	"	"	"				"					"	"	"	"	ENBL to Q0	"	29	"
	t_{PLH4}	"	154	"	"	"	"				"					"	"	"	"	ENBL to Q0	"	39	"
	t_{PLH4}	"	155	4.5 V	"	"	"	OUT			"					"	"	"	"	ENBL to Q1	"	39	"
	t_{PHL4}	"	156	"	"	"	"	"			"					"	"	"	"	ENBL to Q1	"	29	"
	t_{PLH4}	"	157	"	"	"	"	"			"					"	"	"	"	ENBL to Q1	"	39	"
	t_{PLH4}	"	158	GND	4.5 V	"	"	"	OUT		"					"	"	"	"	ENBL to Q2	"	39	"
	t_{PHL4}	"	159	"	"	"	"	"	"		"					"	"	"	"	ENBL to Q2	"	29	"
	t_{PLH4}	"	160	"	"	"	"	"	"		"					"	"	"	"	ENBL to Q2	"	39	"
	t_{PLH4}	"	161	4.5 V	"	"	"	"	"	OUT	"					"	"	"	"	ENBL to Q3	"	39	"
	t_{PHL4}	"	162	"	"	"	"	"	"	"	"					"	"	"	"	ENBL to Q3	"	29	"
	t_{PLH4}	"	163	"	"	"	"	"	"	"	"					"	"	"	"	ENBL to Q3	"	39	"
	t_{PLH4}	"	164	GND	GND	4.5 V	"	"	"	"	"	OUT				"	"	"	"	ENBL to Q4	"	39	"
	t_{PHL4}	"	165	"	"	"	"	"	"	"	"	"				"	"	"	"	ENBL to Q4	"	29	"
	t_{PLH4}	"	166	"	"	"	"	"	"	"	"	"				"	"	"	"	ENBL to Q4	"	39	"
	t_{PLH4}	"	167	4.5 V	"	"	"	"	"	"	"	"	OUT			"	"	"	"	ENBL to Q5	"	39	"
	t_{PHL4}	"	168	"	"	"	"	"	"	"	"	"	"			"	"	"	"	ENBL to Q5	"	29	"
	t_{PLH4}	"	169	"	"	"	"	"	"	"	"	"	"			"	"	"	"	ENBL to Q5	"	39	"
	t_{PLH4}	"	170	GND	4.5 V	"	"	"	"	"	"	"	"	OUT		"	"	"	"	ENBL to Q6	"	39	"
	t_{PHL4}	"	171	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	ENBL to Q6	"	29	"
	t_{PLH4}	"	172	"	"	"	"	"	"	"	"	"	"	"		"	"	"	"	ENBL to Q6	"	39	"
	t_{PLH4}	"	173	4.5 V	"	"	"	"	"	"	"	"	"	"	OUT	"	"	"	"	ENBL to Q7	"	39	"
	t_{PHL4}	"	174	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	ENBL to Q7	"	29	"
	t_{PLH4}	"	175	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	"	ENBL to Q7	"	39	"
	t_{PHL4}	"	176	IN	GND	GND	"	OUT	"	"	"	"	"	"	"	GND	"	"	"	ENBL to Q1	"	29	"
	t_{PLH4}	"	177	"	"	"	"	"	"	"	"	"	"	"	"	4.5 V	"	"	"	ENBL to Q1	"	39	"
	t_{PHL4}	"	178	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	ENBL to Q1	"	29	"
	t_{PHL4}	"	179	GND	IN	"	"	"	OUT	"	"	"	"	"	"	GND	"	"	"	ENBL to Q2	"	29	"
	t_{PLH4}	"	180	"	"	"	"	"	"	"	"	"	"	"	"	4.5 V	"	"	"	ENBL to Q2	"	39	"
	t_{PHL4}	"	181	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	ENBL to Q2	"	29	"
	t_{PHL4}	"	182	"	GND	IN	"	"	"	"	"	OUT	"	"	"	GND	"	"	"	ENBL to Q4	"	29	"
	t_{PLH4}	"	183	"	"	"	"	"	"	"	"	"	"	"	"	4.5 V	"	"	"	ENBL to Q4	"	39	"
	t_{PHL4}	"	184	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	ENBL to Q4	"	29	"
	t_{PHL4}	"	185	IN	"	GND	OUT	"	"	"	"	"	"	"	"	GND	"	"	"	ENBL to Q0	"	29	"
	t_{PLH4}	"	186	"	"	"	"	"	"	"	"	"	"	"	"	4.5 V	"	"	"	ENBL to Q0	"	39	"
	t_{PHL4}	"	187	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	ENBL to Q0	"	29	"
	t_{PHL4}	"	188	4.5 V	IN	4.5 V	"	"	"	"	"	"	OUT	"	"	GND	"	"	"	ENBL to Q5	"	29	"
	t_{PLH4}	"	189	"	"	"	"	"	"	"	"	"	"	"	"	4.5 V	"	"	"	ENBL to Q5	"	39	"
	t_{PHL4}	"	190	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	ENBL to Q5	"	29	"
	t_{PHL4}	"	191	"	4.5 V	IN	"	"	"	OUT	"	"	"	"	"	GND	"	"	"	ENBL to Q3	"	29	"
	t_{PLH4}	"	192	"	"	"	"	"	"	"	"	"	"	"	"	4.5 V	"	"	"	ENBL to Q3	"	39	"
	t_{PHL4}	"	193	"	"	"	"	"	"	"	"	"	"	"	"	GND	"	"	"	ENBL to Q3	"	29	"
10	t_{PHL1}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																			-	42	ns
	t_{PLH2}																				-	48	"
	t_{PHL2}																				-	34	"
	t_{PLH3}																				-	56	"
	t_{PHL3}																				-	44	"
	t_{PLH4}																				-	52	"
	t_{PHL4}																				-	38	"
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F / Cases 2, X / Test no.	1 / 2 / A	2 / 3 / B	3 / 4 / C	4 / 5 / Q0	5 / 7 / Q1	6 / 8 / Q2	7 / 9 / Q3	8 / 10 / GND	9 / 12 / Q4	10 / 13 / Q5	11 / 14 / Q6	12 / 15 / Q7	13 / 17 / DATA IN	14 / 18 / ENBL	15 / 19 / CLR	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
1 T_C = +25°C	V_{OH}	3006	1	0.7 V	0.7 V	0.7 V	-0.4 mA				GND					2.0 V	1/	2.0 V	4.5 V	Q0	2.5		V
			2	2.0 V	0.7 V	"		-0.4 mA			"					"	"	"	"	Q1	"		"
			3	0.7 V	2.0 V	"			-0.4 mA		"					"	"	"	"	Q2	"		"
			4	2.0 V	2.0 V	"				-0.4 mA	"					"	"	"	"	Q3	"		"
			5	0.7 V	0.7 V	2.0 V					"	-0.4 mA				"	"	"	"	Q4	"		"
			6	2.0 V	0.7 V	"					"		-0.4 mA			"	"	"	"	Q5	"		"
			7	0.7 V	2.0 V	"					"			-0.4 mA		"	"	"	"	Q6	"		"
			8	2.0 V	2.0 V	"					"					"	"	"	"	Q7	"		"
	V_{OL}	3007	9	0.7 V	0.7 V	0.7 V	4 mA				"					0.7 V	"	"	"	Q0		0.4	"
			10	2.0 V	0.7 V	"		4 mA			"					"	"	"	"	Q1		"	"
			11	0.7 V	2.0 V	"			4 mA		"					"	"	"	"	Q2		"	"
			12	2.0 V	2.0 V	"				4 mA	"					"	"	"	"	Q3		"	"
			13	0.7 V	0.7 V	2.0 V					"	4 mA				"	"	"	"	Q4		"	"
			14	2.0 V	0.7 V	"					"		4 mA			"	"	"	"	Q5		"	"
			15	0.7 V	2.0 V	"					"			4 mA		"	"	"	"	Q6		"	"
			16	2.0 V	2.0 V	"					"					"	"	"	"	Q7		"	"
	I_{IL1}	3009	17	0.4 V															5.5 V	A	4/	4/	mA
			18		0.4 V	0.4 V										0.4 V				B			
			19														0.4 V			C			
			20																	D1			
			21														0.4 V			ENBL			
			22															0.4 V		CLR			
	I_{IH1}	3010	23	2.7 V																A		20	μA
			24		2.7 V	2.7 V										2.7 V				B			
			25														2.7 V			C			
			26																	D1			
			27														2.7 V			ENBL			
			28															2.7 V		CLR			
	I_{IH3}		29	5.5 V																A		100	μA
			30		5.5 V	5.5 V										5.5 V				B			
			31														5.5 V			C			
			32																	D1			
			33														5.5 V			ENBL			
			34															5.5 V		CLR			
	V_{IC}		35	-18 mA															4.5 V	A		-1.5	V
			36		-18 mA	-18 mA										-18 mA				B			
			37														-18 mA			C			
			38																	D1			
			39														-18 mA			ENBL			
			40															-18 mA		CLR			
	I_{OS}	3011	41	GND	GND	GND	GND	GND	GND	GND		GND	GND	GND	GND	5.0 V	GND	5.0 V	5.5 V	Q0	-15	-100	mA
			42	5.0 V	GND	"										"	"	"	"	Q1	"	"	"
			43	GND	5.0 V	"										"	"	"	"	Q2	"	"	"
			44	5.0 V	5.0 V	"										"	"	"	"	Q3	"	"	"
			45	GND	GND	5.0 V										"	"	"	"	Q4	"	"	"
			46	5.0 V	GND	"										"	"	"	"	Q5	"	"	"
			47	GND	5.0 V	"										"	"	"	"	Q6	"	"	"
			48	5.0 V	5.0 V	"										"	"	"	"	Q7	"	"	"
	I_{CC}	3005	49	GND	GND	GND					"					GND	"	GND	"	V_{LC}		36	"

See footnotes at end of device type 05.

MIL-M-38510/316D

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases E, F	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Measured terminal	Limits		Unit
			Cases 2, X	2	3	4	5	7	8	9	10	12	13	14	15	17	18	19	20				
			Test no.	A	B	C	Q0	Q1	Q2	Q3	GND	Q4	Q5	Q6	Q7	DATA IN	ENBL	CLR	V_{CC}		Min	Max	
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																						
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																						
7 2/ T_C = +25°C	Truth table tests	3014	50	B	B	B	L	L	L	L	GND	L	L	L	L	A	A	B	5.0 V	3/			
			51	B	A	-	H	-	-	-		-	-	-	-	-	B	-					
			52	A	A	-	L	-	-	-		-	-	-	-	-	A	-					
			53	A	B	-	-	H	-	-		-	-	-	-	-	B	-					
			54	B	B	A	-	L	-	-		-	-	-	-	-	A	-					
			55	B	A	-	-	-	H	-		-	-	-	-	-	B	-					
			56	A	A	-	-	-	L	-		-	-	-	-	-	A	-					
			57	A	B	B	-	-	-	H		-	-	-	-	-	B	-					
			58	B	B	-	-	-	-	L		-	-	-	-	-	A	-					
			59	B	A	A	-	-	-	-		H	-	-	-	-	B	-					
			60	A	A	-	-	-	-	-		L	-	-	-	-	A	-					
			61	A	B	A	-	-	-	-		-	H	-	-	-	B	-					
			62	B	B	-	-	-	-	-		-	L	-	-	-	A	-					
			63	B	A	-	-	-	-	-		-	-	H	-	-	B	-					
			64	A	-	-	-	-	-	-		-	-	L	-	-	A	-					
			65	-	B	B	-	-	-	-		-	-	-	H	-	B	-					
			66	-	A	A	-	-	-	-		-	-	-	L	-	A	-					
			67	B	B	-	-	-	-	-		-	-	-	-	-	B	-					
			68	A	A	-	-	-	-	-		-	-	-	-	-	A	-					
			69	B	-	B	-	-	-	-		-	-	-	-	-	B	-					
			70	A	-	A	-	-	-	-		-	-	-	-	-	A	-					
			71	B	-	A	-	-	-	-		-	-	-	-	-	B	-					
			72	A	-	B	-	-	-	-		-	-	-	-	-	A	-					
			73	B	-	-	-	-	-	-		-	-	-	-	-	B	-					
			74	-	-	-	H	-	-	-		-	-	-	-	-	A	-					
			75	-	A	-	-	-	-	-		-	-	-	-	-	B	-					
			76	-	-	-	-	-	-	-		-	-	-	-	-	A	-					
			77	-	B	-	-	H	-	-		-	-	-	-	-	B	-					
			78	-	-	-	-	-	-	-		-	-	-	-	-	A	-					
			79	-	A	-	-	-	-	-		-	-	-	-	-	B	-					
			80	-	-	-	-	-	H	-		-	-	-	-	-	A	-					
			81	-	B	-	-	-	-	-		-	-	-	-	-	B	-					
			82	-	A	-	-	-	-	-		-	-	-	-	-	A	-					
			83	-	-	-	-	-	-	-		-	-	-	-	-	B	-					
			84	-	-	-	-	-	-	H		-	-	-	-	-	A	-					
			85	-	B	B	-	-	-	-		-	-	-	-	-	B	-					
			86	-	-	-	A	-	-	-		H	-	-	-	-	A	-					
			87	-	A	-	-	-	-	-		-	-	-	-	-	B	-					
			88	-	-	-	-	-	-	-		-	-	-	-	-	A	-					
			89	-	B	-	-	-	-	-		-	H	-	-	-	B	-					
			90	-	-	-	-	-	-	-		-	-	-	-	-	A	-					
			91	-	A	A	-	-	-	-		-	-	-	-	-	B	-					
			92	-	-	-	-	-	-	-		-	-	-	-	-	A	-					
			93	-	-	-	-	-	-	-		-	-	-	-	-	B	-					
			94	-	A	-	-	-	-	-		-	-	-	-	-	A	-					
			95	-	A	-	-	-	-	-		-	-	-	-	-	B	-					
8	Same tests and terminal conditions as for subgroup 7, except T_C = -55°C and +125°C.																						

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 method	Cases L, F Cases 2, X Test no.	1 2 A	2 3 B	3 4 C	4 5 Q0	5 7 Q1	6 8 Q2	7 9 Q3	8 10 GND	9 12 Q4	10 13 Q5	11 14 Q6	12 15 Q7	13 17 DATA IN	14 18 ENBL	15 19 CLR	16 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PHL1}	3003 Fig. 7	96	GND	GND	GND	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	4.5 V	0/	IN	5.0 V	CLR to Q0	3	23	ns
			97	4.5 V	GND	"					"					"	"	"	"	CLR to Q1	"	"	"
			98	GND	4.5 V	"					"					"	"	"	"	CLR to Q2	"	"	"
			99	4.5 V	4.5 V	"					"					"	"	"	"	CLR to Q3	"	"	"
			100	GND	GND	4.5 V					"					"	"	"	"	CLR to Q4	"	"	"
			101	4.5 V	GND	"					"					"	"	"	"	CLR to Q5	"	"	"
			102	GND	4.5 V	"					"					"	"	"	"	CLR to Q6	"	"	"
			103	4.5 V	4.5 V	"					"					"	"	"	"	CLR to Q7	"	"	"
	t_{PLH2}	"	104	GND	GND	GND	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	IN	GND	GND	"	D1 to Q0	"	35	"
			105	4.5 V	GND	"					"					"	"	"	"	D1 to Q1	"	"	"
			106	GND	4.5 V	"					"					"	"	"	"	D1 to Q2	"	"	"
			107	4.5 V	4.5 V	"					"					"	"	"	"	D1 to Q3	"	"	"
			108	GND	GND	4.5 V					"					"	"	"	"	D1 to Q4	"	"	"
			109	4.5 V	GND	"					"					"	"	"	"	D1 to Q5	"	"	"
			110	GND	4.5 V	"					"					"	"	"	"	D1 to Q6	"	"	"
			111	4.5 V	4.5 V	"					"					"	"	"	"	D1 to Q7	"	"	"
	t_{PHL2}	"	112	GND	GND	GND	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	"	"	"	"	D1 to Q0	"	25	"
			113	4.5 V	GND	"					"					"	"	"	"	D1 to Q1	"	"	"
			114	GND	4.5 V	"					"					"	"	"	"	D1 to Q2	"	"	"
			115	4.5 V	4.5 V	"					"					"	"	"	"	D1 to Q3	"	"	"
			116	GND	GND	4.5 V					"					"	"	"	"	D1 to Q4	"	"	"
			117	4.5 V	GND	"					"					"	"	"	"	D1 to Q5	"	"	"
			118	GND	4.5 V	"					"					"	"	"	"	D1 to Q6	"	"	"
			119	4.5 V	4.5 V	"					"					"	"	"	"	D1 to Q7	"	"	"
	t_{PLH3}	"	120	IN	GND	GND	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	4.5 V	"	"	"	A to Q0	"	32	"
			121	IN	GND	"					"					"	"	"	"	A to Q1	"	"	"
			122	GND	IN	"					"					"	"	"	"	B to Q2	"	"	"
			123	IN	4.5 V	"					"					"	"	"	"	A to Q3	"	"	"
			124	GND	GND	IN					"					"	"	"	"	C to Q4	"	"	"
			125	IN	GND	4.5 V					"					"	"	"	"	A to Q5	"	"	"
			126	GND	IN	"					"					"	"	"	"	B to Q6	"	"	"
			127	IN	4.5 V	"					"					"	"	"	"	A to Q7	"	"	"
	t_{PHL3}	"	128	IN	GND	GND	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	"	"	"	"	A to Q0	"	"	"
			129	IN	GND	"					"					"	"	"	"	A to Q1	"	"	"
			130	GND	IN	"					"					"	"	"	"	B to Q2	"	"	"
			131	IN	4.5 V	"					"					"	"	"	"	A to Q3	"	"	"
			132	GND	GND	IN					"					"	"	"	"	C to Q4	"	"	"
			133	IN	GND	4.5 V					"					"	"	"	"	A to Q5	"	"	"
			134	GND	IN	"					"					"	"	"	"	B to Q6	"	"	"
			135	IN	4.5 V	"					"					"	"	"	"	A to Q7	"	"	"
	t_{PLH4}	"	136	GND	GND	GND	OUT	OUT	OUT	OUT	"	OUT	OUT	OUT	OUT	"	IN	"	"	ENBL to Q0	"	29	"
			137	4.5 V	GND	"					"					"	"	"	"	ENBL to Q1	"	"	"
			138	GND	4.5 V	"					"					"	"	"	"	ENBL to Q2	"	"	"
			139	4.5 V	4.5 V	"					"					"	"	"	"	ENBL to Q3	"	"	"
			140	GND	GND	4.5 V					"					"	"	"	"	ENBL to Q4	"	"	"
			141	4.5 V	GND	"					"					"	"	"	"	ENBL to Q5	"	"	"
			142	GND	4.5 V	"					"					"	"	"	"	ENBL to Q6	"	"	"
			143	4.5 V	4.5 V	"					"					"	"	"	"	ENBL to Q7	"	"	"

See footnotes at end of device type 05.

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TABLE III. Group A inspection for device type 05 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V or low ≤0.7 V or open).

Subgroup	Symbol	MIL-STD-883 Method	Cases E, F / Cases 2, X Test no.	1 / 2 / A	2 / 3 / B	3 / 4 / C	4 / 5 / Q0	5 / 7 / Q1	6 / 8 / Q2	7 / 9 / Q3	8 / 10 / GND	9 / 12 / Q4	10 / 13 / Q5	11 / 14 / Q6	12 / 15 / Q7	13 / 17 / DATA IN	14 / 18 / ENBL	15 / 19 / CLR	16 / 20 / V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PHL4}	3003 Fig. 7	144	GND	GND	GND	OUT	OUT	OUT	OUT	GND	OUT	OUT	OUT	OUT	4.5 V	IN	GND	5.0 V	ENBL to Q0	3	29	ns
			145	4.5 V	GND	"					"					"	"	"	"	ENBL to Q1	"	"	"
			146	GND	4.5 V	"					"					"	"	"	"	ENBL to Q2	"	"	"
			147	4.5 V	4.5 V	"					"					"	"	"	"	ENBL to Q3	"	"	"
			148	GND	GND	4.5 V					"					"	"	"	"	ENBL to Q4	"	"	"
			149	4.5 V	GND	"					"					"	"	"	"	ENBL to Q5	"	"	"
			150	GND	4.5 V	"					"					"	"	"	"	ENBL to Q6	"	"	"
			151	4.5 V	4.5 V	"					"					"	"	"	"	ENBL to Q7	"	"	"
	t_{PLH4}	5/	152	GND	GND	GND	OUT				"					IN	IN	4.5 V	"	ENBL to Q0	"	"	"
	t_{PHL4}		153	"	"	"	"				"					"	"	"	"	ENBL to Q0	"	"	"
	t_{PLH4}		154	"	"	"					"					"	"	"	"	ENBL to Q0	"	"	"
	t_{PLH4}		155	4.5 V	"	"		OUT			"					"	"	"	"	ENBL to Q1	"	"	"
	t_{PHL4}		156	"	"	"		"			"					"	"	"	"	ENBL to Q1	"	"	"
	t_{PLH4}		157	"	"	"					"					"	"	"	"	ENBL to Q1	"	"	"
	t_{PLH4}		158	GND	4.5 V	"			OUT		"					"	"	"	"	ENBL to Q2	"	"	"
	t_{PHL4}		159	"	"	"			"		"					"	"	"	"	ENBL to Q2	"	"	"
	t_{PLH4}		160	"	"	"					"					"	"	"	"	ENBL to Q2	"	"	"
	t_{PLH4}		161	4.5 V	"	"				OUT	"					"	"	"	"	ENBL to Q3	"	"	"
	t_{PHL4}		162	"	"	"				"	"					"	"	"	"	ENBL to Q3	"	"	"
	t_{PLH4}		163	"	"	"					"					"	"	"	"	ENBL to Q3	"	"	"
	t_{PLH4}		164	GND	GND	4.5 V					"	OUT				"	"	"	"	ENBL to Q4	"	"	"
	t_{PHL4}		165	"	"	"					"	"				"	"	"	"	ENBL to Q4	"	"	"
	t_{PLH4}		166	"	"	"					"					"	"	"	"	ENBL to Q4	"	"	"
	t_{PLH4}		167	4.5 V	"	"					"		OUT			"	"	"	"	ENBL to Q5	"	"	"
	t_{PHL4}		168	"	"	"					"		"			"	"	"	"	ENBL to Q5	"	"	"
	t_{PLH4}		169	"	"	"					"					"	"	"	"	ENBL to Q5	"	"	"
	t_{PLH4}		170	GND	4.5 V	"					"			OUT		"	"	"	"	ENBL to Q6	"	"	"
	t_{PHL4}		171	"	"	"					"			"		"	"	"	"	ENBL to Q6	"	"	"
	t_{PLH4}		172	"	"	"					"					"	"	"	"	ENBL to Q6	"	"	"
	t_{PLH4}		173	4.5 V	"	"					"				OUT	"	"	"	"	ENBL to Q7	"	"	"
	t_{PHL4}		174	"	"	"					"				"	"	"	"	"	ENBL to Q7	"	"	"
	t_{PLH4}		175	"	"	"					"					"	"	"	"	ENBL to Q7	"	"	"
	t_{PHL4}		176	IN	GND	GND		OUT			"					GND	"	"	"	ENBL to Q1	"	"	"
	t_{PLH4}		177	"	"	"		"			"					4.5 V	"	"	"	ENBL to Q1	"	"	"
	t_{PHL4}		178	"	"	"					"					GND	"	"	"	ENBL to Q1	"	"	"
	t_{PHL4}		179	GND	IN	"			OUT		"					GND	"	"	"	ENBL to Q2	"	"	"
	t_{PLH4}		180	"	"	"			"		"					4.5 V	"	"	"	ENBL to Q2	"	"	"
	t_{PHL4}		181	"	"	"					"					GND	"	"	"	ENBL to Q2	"	"	"
	t_{PHL4}		182	"	GND	IN					"	OUT				GND	"	"	"	ENBL to Q4	"	"	"
	t_{PLH4}		183	"	"	"					"	"				4.5 V	"	"	"	ENBL to Q4	"	"	"
	t_{PHL4}		184	"	"	"					"					GND	"	"	"	ENBL to Q4	"	"	"
	t_{PHL4}		185	IN	"	GND	OUT				"					GND	"	"	"	ENBL to Q0	"	"	"
	t_{PLH4}		186	"	"	"	"				"					4.5 V	"	"	"	ENBL to Q0	"	"	"
	t_{PHL4}		187	"	"	"					"					GND	"	"	"	ENBL to Q0	"	"	"
	t_{PHL4}		188	4.5 V	IN	4.5 V					"		OUT			GND	"	"	"	ENBL to Q5	"	"	"
	t_{PLH4}		189	"	"	"					"		"			4.5 V	"	"	"	ENBL to Q5	"	"	"
	t_{PHL4}		190	"	"	"					"					GND	"	"	"	ENBL to Q5	"	"	"
	t_{PHL4}		191	"	4.5 V	IN				OUT	"					GND	"	"	"	ENBL to Q3	"	"	"
	t_{PLH4}		192	"	"	"				"	"					4.5 V	"	"	"	ENBL to Q3	"	"	"
	t_{PHL4}		193	"	"	"					"					GND	"	"	"	ENBL to Q3	"	"	"
10	t_{PHL1}, t_{PLH2}, t_{PHL2}, t_{PLH3}, t_{PHL3}, t_{PLH4}, t_{PHL4}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits are as shown.																					
11	Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																						

See footnotes at end of device type 05.

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Device types 03 and 05

1/ Apply a 3V/0V/3V momentary pulse 500 ns minimum prior to measurement.

2/ A = 2.4 V, B = 0.4 V.

3/ H $\geq$ 1.5 V, L $\leq$ 1.5 V.

4/ I_{IL} limits are as follows:

Symbol	Min/max limits (mA)	
	Circuit B, C Device 03	Circuit A Device 05
IIL1	-.12/-.36	-.16/-.40
	Test 21 -.005/-.72	Test 21 -.005/-0.20

5/ See note 6 of figure 7.

6/ Apply 4.5 V / 0 V momentary pulse prior to each test.

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4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical test requirements shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions.

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

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6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential
V_{IN}	Voltage level at an input terminal
I_{IN}	Current flowing into an input terminal
V_{SUH}	Setup high
V_{SUL}	Setup low

6.4 Logistic support. Lead materials and finishes (see 3.3), are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer lead lengths and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510. Device 05 is substitutable for 03 when t_{SETUP} and t_{HOLD} are not crucial. When a faster propagation delay time is needed use device 05.

Military device type	Generic-industry type
01	54LS75
02	54LS279
03	54LS259
04	54LS375
05	54LS259B

6.6 Manufacturers' designation. Manufacturers' circuits which form a part of this specification are designated with an "X" as shown in table IV herein.

TABLE IV. Manufacturers' designation.

Device type	Circuits				
	A	B	C	D	E
	Texas Instruments	Motorola	Fairchild	Signetics	National Semiconductor
01	X	X		X	X
02	X	X	X	X	X
03	X	X	X		X
04	X	X	X	X	
05	X				

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6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1077)

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